

Check-In 2

Group Name: HeyBene

Calerb Louis Jean, Haoran Huo, Xinqi Tao

cjean33@gatech.edu, hhuo30@gatech.edu,

taoxinqi23@gatech.edu

1 INTRODUCTION

For this check-in, we are sharing our needfinding results along with the three design alternatives we came up with after our brainstorming session. We collected 20 survey responses and ran 4 semi-structured interviews, and honestly the data pointed us in a pretty clear direction. Our goal from the start was to move the OpenBot (Müller and Koltun, 2021) control experience away from what it currently feels like, an engineering debug screen toward something that actually feels natural to hold and use. The findings gave us a solid foundation for that.

2 NEEDFINDING RESULTS

We sent the survey out to a mix of people. Some with prior experience controlling RC cars or robots, some completely new to it. We ended up with 20 responses total. Two participants did not complete the full survey, so for questions 5 through 8 we are working with 18 responses instead of 20. We made sure to design the survey to avoid the usual pitfalls, no yes/no questions and no leading phrasing, so the answers would reflect what people actually think rather than what they thought we wanted to hear.

A significant majority of our respondents (85%) had prior experience with physical gamepads or remotes, which gave us a really strong baseline for thinking about mental models. Combining the survey data with the interview feedback, we pulled out three insights that kept coming up over and over again.

2.1 Insight 1: The Video Feed and Responsiveness Come Before Everything Else

We actually went into this assuming that users would be overwhelmed by too much technical data on screen. The data kind of flipped that assumption. What people care about most is not hiding information it is making sure the app feels fast and gives them a clear window into what the car is seeing.

- **Data:** 80% of respondents (16 out of 20) selected the live camera feed as their top required element on screen. Speed and battery followed at 55% each (11 out

of 20). But the really telling number was in the frustrations question 14 out of 18 respondents who answered that question, or 78%, said that controls feeling delayed or laggy was their biggest pain point.

In our interviews, one participant who flies drones put it pretty directly:

"If the video feed stutters, the robot is practically undriveable, no matter how good the UI looks."

What this means for our design: The interface has to be built around a low-latency, unobstructed camera feed first. Essential info like speed and battery should be there as clean overlays in the corners, not taking up dedicated screen space. And the secondary engineering stuff motor parameters, PID values needs to be tucked away somewhere and not shown during normal driving. About 39% of respondents still said too much information on screen bothered them, so even though people tolerate some data, clutter is still a real concern.

2.2 Insight 2: Thumbsticks Feel Natural, But the Tactile Gap is a Real Problem

Since OpenBot replaces a physical controller with a phone screen, we really needed to understand what control mental models people bring with them. The answer was pretty clear, but with an important catch.

- **Data:** 55% of respondents (11 out of 20) said that two on-screen thumbsticks like a gamepad would feel most natural to them. Tilt-to-steer came in second at 20% (4 out of 20), arrow buttons at 15% (3 out of 20), and the remaining two respondents preferred other options like a single thumbstick with a stop button or a steering wheel slider.

So thumbsticks are clearly the dominant preference. But the frustration data complicates that a bit 39% of respondents (7 out of 18) said it was hard to tell whether the device was actually responding to their input. In the interviews, one mobile gamer said:

"I like the joysticks on the screen, but because I can't physically feel them, I sometimes slide my thumbs completely out of the zone."

What this means for our design: We cannot just scrap the thumbstick layout since most people prefer it, but we also cannot just use it as-is. The key challenge is bridging that tactile gap making the input zones large and forgiving, adding visual or haptic feedback so users actually know their input is registering, and making it so that drifting slightly off the joystick area does not just stop the car cold.

2.3 Insight 3: Mode Switching Needs to Be Impossible to Miss

Switching between manual driving and the auto-tracking feature is one of the most important actions in the app, and right now there is no clear or consistent way to do it. The data told us exactly what users expect.

- **Data:** Of the 18 respondents who answered this question, 10 (56%) preferred a clearly labeled toggle switch for switching modes. Camera-style left/right swipe came in second at 28% (5 out of 18), followed by tabs at the top at 22% (4 out of 18). And when we asked what would make the switch feel clear and easy, 61% said the current mode should always be visible on screen, and 50% emphasized simple and clear labels.

One of our interviewees put it perfectly:

"I do not want to dig into a menu. It should be one big button, and the screen should make it obvious who is driving."

What this means for our design: Mode switching cannot be buried. It has to be a primary action that is always one tap away, with the active mode persistently visible on the HUD. Color-coding the interface based on the active mode seems like a really straightforward way to give users that constant reassurance.

3 INITIAL BRAINSTORMING PLAN

After we all reviewed the needfinding results together, we put together a structured plan for generating design alternatives. We wanted to make sure we avoided two specific traps from the readings: production blocking, where people forget their ideas while waiting to share them, and anchoring bias, where the first idea thrown out ends up dominating the whole session. To get around those, we used a two-step approach:

1. **Asynchronous brain-writing first:** Each team member spent about 24 hours independently sketching out raw interface concepts on paper before we got together. The rule was to defer judgment completely and just generate as many ideas as possible, even weird ones. We focused specifically on ideas that addressed our three insights visual primacy, the tactile gap, and explicit mode switching.
2. **Synchronous affinity mapping second:** We then met up and pooled everything onto a shared whiteboard. We grouped ideas into functional buckets like "Steer-

ing Paradigms," "HUD Overlays," and "Autonomy Toggles," and then looked at which clusters had the most overlap with what users actually said they wanted.

4 BRAINSTORMING RESULTS AND DESIGN ALTERNATIVES

Across our session we came up with over 15 individual feature ideas, ranging from voice-controlled navigation to AR path overlays. After running them through our three insights as a filter, we cut anything that was either too complex to realistically implement or that went against the minimalist, low-latency experience our data pointed to. We ended up landing on three distinct design alternatives, each with a different philosophy for solving the tactile gap while keeping the camera feed front and center.

4.1 Alternative 1: The "Mobile Gamer" Layout

Rationale: This one is built directly for the 55% of users who wanted thumbstick controls. Instead of ditching the familiar gamepad mental model, the idea is to fix its biggest touchscreen weakness with a smarter implementation.

Prototype Description: The phone is held horizontally. The whole screen is the live camera feed. To deal with the drifting problem our interviewees mentioned, the left joystick is "floating" it snaps to wherever the user's left thumb initially lands on the glass instead of being fixed in one spot. The right side has a large pill-shaped acceleration pad. At the top center there is a prominent toggle switch between "Manual" and "Auto Tracking" that turns bright green when the car is in auto mode, so the current state is always visible at a glance.

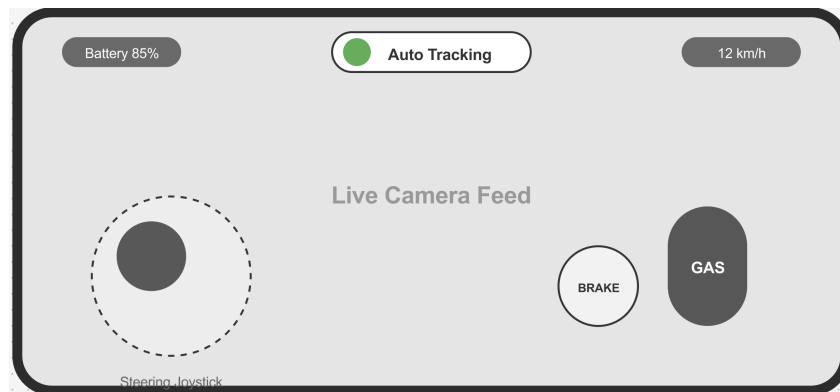


Figure 1—Wireframe prototype of Alternative 1: The Mobile Gamer layout with floating thumbstick.

4.2 Alternative 2: The "Immersive FPV" (Tilt-to-Steer)

Rationale: This one is for the 20% who preferred tilt controls and the 39% who were frustrated by screen clutter. It removes on-screen directional controls entirely, which clears the view and leans into the Match Between System and the Real-World heuristic, tilting a phone to steer is physically similar to turning a steering wheel.

Prototype Description: The phone is held horizontally and the gyroscope handles all steering tilt left or right to turn. A single vertical slider on the right edge controls speed and braking. With no joystick overlays blocking the view, almost the entire screen is the live First-Person View camera feed. Battery and speed are displayed as small semi-transparent overlays in the top corners, keeping the essential info present without cluttering the view.

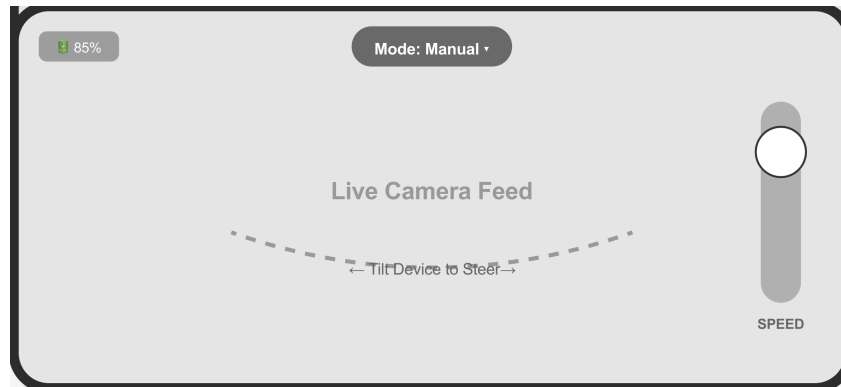


Figure 2—Wireframe prototype of Alternative 2: Immersive FPV layout utilizing tilt-to-steer.

4.3 Alternative 3: The "One-Handed Casual" (Vertical)

Rationale: This one came from our interview with a participant who just wanted something they could use casually with one hand while walking, without the two-handed commitment of a traditional controller layout.

Prototype Description: The phone is held vertically in portrait mode. The top half of the screen is the live camera feed with a persistent mode banner right above it. The bottom half is a clean semi-circular swipe pad, the user places their thumb on it and drags up to accelerate or slides left/right to steer. Releasing the thumb triggers an auto-brake. The whole thing is one point of contact, which means the user barely needs to look at the screen at all once they get the feel for it.

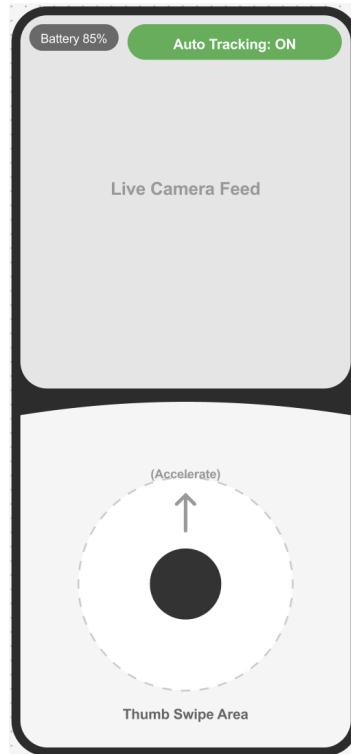


Figure 3—Wireframe prototype of Alternative 3: The One-Handed Casual vertical layout.

5 FEEDBACK WE WANT

As we move into medium-fidelity prototyping, the main things we would love feedback on are:

- Which of the three alternatives seems most promising for a general audience, someone who has never touched an OpenBot before?
- Do the prototype descriptions actually address the tactile gap issue, or is there a gap we are still not solving?
- Are there any critical status indicators or safety cues we are missing that should stay visible in a minimalist design?

REFERENCES

1. Nielsen, Jakob (1994). “10 Usability Heuristics for User Interface Design”. In: *Nielsen Norman Group*. URL: <https://www.nngroup.com/articles/ten-usability-heuristics/>.

2. Delong, M. L. et al. (2020). "Review of Human–Machine Interfaces for Small Unmanned Systems With Robotic Manipulators". In: *IEEE Transactions on Human-Machine Systems* 50.3, pp. 250–264.
3. Müller, Matthias and Koltun, Vladlen (2021). "OpenBot: Turning Smartphones into Robots". In: *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, pp. 5003–5009.

A NEEDFINDING RAW DATA

The survey consisted of 8 multiple-choice questions distributed to 20 participants. Two participants did not complete questions 5 through 8, so the effective sample size for those questions is 18. Percentages for Q1–Q4 are out of N=20; percentages for Q5–Q8 are out of N=18.

Survey Results

Q1: Have you ever controlled a remote-control car, drone, or robot before?

- Yes — with a physical gamepad or remote: 13 (65%)
- Yes — with both: 4 (20%)
- Yes — with a phone app: 2 (10%)
- No, never: 1 (5%)

Q2: How comfortable would you feel using your phone as a controller? (Scale 0–10)

Average Score: 5.85 / 10

Q3: Which input method would feel most natural? (phone held horizontally)

- Two on-screen thumbsticks like a gamepad: 11 (55%)
- Tilt the phone itself to steer: 4 (20%)
- On-screen arrow buttons: 3 (15%)
- Single steering wheel slider or other: 2 (10%)

Q4: What information would you most want to see on screen at all times? (select all that apply)

- Live camera feed from the car: 16 (80%)
- Speed/motor output: 11 (55%)
- Battery level: 11 (55%)
- Connection status: 8 (40%)
- Current drive mode: 6 (30%)

Q5: Comfort with technical settings on screen during normal driving (Scale 0–10, N=17)

Average Score: 5.76 / 10 (3 participants skipped this question)

Q6: What do you find most frustrating about controlling physical devices with your phone? (N=18)

- Controls feel delayed / laggy: 14 (78%)
- Too much information on the screen: 7 (39%)
- Hard to tell whether the device is responding: 7 (39%)
- Connection issues: 6 (33%)
- Interface is hard to understand at first: 4 (22%)

Q7: How would you expect to switch between manual driving and automatic tracking mode? (N=18)

- A clearly labeled toggle switch: 10 (56%)
- Swipe left/right like a camera app: 5 (28%)
- Tabs at the top of the screen: 4 (22%)
- Two separate buttons for the two modes: 2 (11%)
- A pop-up menu: 2 (11%)

Q8: What would make that mode switch feel clear and easy? (N=18)

- The current mode is always visible: 11 (61%)
- The labels are simple and clear: 9 (50%)
- The app gives visual feedback after switching: 5 (28%)

Interview Highlights

- **"The Tech Geek" (Prior OpenBot Experience):** "I tried the phone app but the delay drives me crazy. With a physical remote I have muscle memory. On a screen my thumbs drift off the joysticks because I can't feel them. Keep the video feed large I don't mind speed or battery in the corners."
- **"The Drone Pilot" (RC Experience):** "The biggest issue with phone controllers is the lag. If the video stutters, you crash. Also the screen should not be cluttered with useless data. Mode switching should just be a toggle. I need to know instantly if I'm flying it or if the AI is flying it."
- **"The Mobile Gamer" (Novice):** "Definitely two thumbsticks. Left for movement, right for steering, it's what everyone is used to. Games give you a little visual ripple when you press a button. Without that it's hard to tell if the car is actually receiving my command."

- **"The Minimalist" (Novice, Non-Gamer):** "I'd love it if I could just swipe left or right to change modes, like swiping from Photo to Video on an iPhone. And just simple labels: Auto Mode or Manual Mode right at the top so I don't get confused."